

Capstone Project and Test Summary

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Playwright JavaScript Automation

Demo Web Shop E-Commerce Testing

Field	Project Value
Application Under Test	Demo Web Shop
Website URL	https://demowebshop.tricentis.com/
Automation Tool	Playwright Test with JavaScript
Automation Design	Page Object Model with service-wise spec folders
Final Services Covered	8 services
Final Test Cases	120 unique test cases
Cross-Browser Executions	360 executions: Chromium, Firefox, WebKit
Final Result	360 passed, 0 failed, 0 skipped
Hosted Allure Report	https://ajaybhatnagar1712.github.io/Wipro_SDET_Playwright_Capstone/

Overview

The Application Under Test (AUT) is Demo Web Shop, a public e-commerce demo application used to validate real customer-facing workflows such as authentication, catalog browsing, cart validation, account access, shipping information, checkout visibility, wishlist/compare behavior, and customer support pages.

This document summarizes the final Playwright JavaScript capstone project, including the testing goals, functional scope, automation design, service-wise test coverage, CI/CD workflow, and final Allure execution result. The project follows a Page Object Model structure and executes the same functional tests on Chromium, Firefox, and WebKit.

Goals

1 Business Objectives

- Validate stable customer journeys on a public e-commerce application without depending on developer-only access.
- Verify that major shopping workflows expose the expected pages, controls, labels, and navigation paths.
- Provide a professional evidence trail through GitHub Actions and a hosted Allure report.

2 Quality Engineering Objectives

- Automate at least 120 functional test cases across 8 services.
- Run the complete suite on Chromium, Firefox, and WebKit using Playwright projects.
- Maintain readable test names, service-wise folders, and reusable Page Object classes.
- Generate a combined Allure report and deploy it through GitHub Pages.

Specifications

In-Scope Functional Modules

Module	Requirement Description	Priority
Authentication	Login, registration, password recovery, protected guest redirects, header account links	High
Product and Search	Home page, categories, product details, search results, sorting controls	High
Cart	Empty cart, cart navigation, add-to-cart controls, checkout journey labels	High
Wishlist and Compare	Wishlist page, compare page, product wishlist/compare actions, email-a-friend pages	Medium
User Profile and Account	Account, orders, addresses, register/login account links and sitemap account entries	High

Address and Shipping	Shipping/returns pages, shipping text, availability labels, checkout progress steps	Medium
Checkout and Payment	Cart checkout guards, payment/confirm/complete steps, gift card fields and price labels	High
Customer Support and Information	Contact us, sitemap, privacy notice, conditions, about us, news, blog, footer links	Medium

Milestone 1 - Analysis and Planning

1.1 Introduction

The planning phase defined the services to be covered, the website stability constraints, and the automation approach. Demo Web Shop was selected because it is publicly accessible, does not require private backend access, and has enough e-commerce workflows to satisfy the capstone coverage requirement.

1.2 Requirement Analysis

Requirement ID	Requirement Summary	Mapped Service
FR-01	Users should be able to access login, registration, and recovery pages.	Authentication
FR-02	Catalog categories and product detail pages should render correctly.	Product and Search
FR-03	Cart pages and add-to-cart controls should remain visible and usable.	Cart
FR-04	Wishlist, compare, and email-a-friend flows should expose expected controls.	Wishlist and Compare
FR-05	Guest users should be redirected to login for protected account pages.	User Profile and Account
FR-06	Shipping pages and checkout progress labels should be visible.	Address and Shipping
FR-07	Checkout and payment-related controls should be available or guarded correctly.	Checkout and Payment
FR-08	Information and customer support pages should be reachable.	Customer Support and Information

1.3 Test Strategy

- Service-wise test organization: each major business area has its own folder under tests.
- Page Object Model: page classes store reusable navigation and assertion methods.
- Cross-browser coverage: Playwright projects run the same suite on Chromium, Firefox, and WebKit.

- Stable guest-user validations: the suite avoids risky repeated account creation or real payment submission.
- Allure reporting: each CI run produces an HTML report and deploys it to GitHub Pages.

Milestone 2 - Project Setup and Static Review

2.1 Folder Structure

The final project structure is organized for maintainability and service-level reporting. The files below are the core source folders used by the test suite:

Folder	Purpose
.github/workflows	GitHub Actions workflow for Playwright execution and Allure deployment
api	Reserved helper layer for API/client utility structure
data	Project data and expected coverage constants
fixtures	Reusable fixture exports for page objects
pages	Page Object Model classes for each service
tests	Service-wise Playwright spec files
docs	Training notes, test plan, schedule, and final report artifacts

2.2 Dependencies and Tooling

Dependency / Tool	Usage
@playwright/test	Core Playwright test runner and assertion library
allure-playwright	Allure reporter integration for Playwright
allure-commandline	Allure HTML report generation and local report opening
GitHub Actions	CI pipeline for automated test execution
GitHub Pages	Static hosting for the generated Allure report

Milestone 3 - Playwright Automation Testing

3.1 Playwright Configuration

The project uses Playwright Test with one worker, retry enabled on CI, and three browser projects. The main browser matrix is:

Browser Project	Execution Purpose
chromium	Chrome-like browser validation
firefox	Firefox browser compatibility validation
webkit	Safari/WebKit compatibility validation

```
projects: [  
  { name: 'chromium', use: { ...devices['Desktop Chrome'] } },  
  { name: 'firefox', use: { ...devices['Desktop Firefox'] } },  
  { name: 'webkit', use: { ...devices['Desktop Safari'] } },  
]
```

3.2 Automation Design Pattern

Each test file imports the matching Page Object class, creates the page object in test.beforeEach, and calls a readable service method. This keeps spec files small and makes the test case names clear in Playwright HTML and Allure reports.

```
import { test } from '@playwright/test';  
import { AuthenticationPage } from '../pages/authentication.page.js';  
  
test.describe('Authentication Module', () => {  
  let authenticationPage;  
  test.beforeEach(async ({ page }) => {  
    authenticationPage = new AuthenticationPage(page);  
  });  
});
```

3.3 Test Case Coverage Summary

Service	Test Cases	Status
Authentication	15	Implemented
Product and Search	15	Implemented
Cart	15	Implemented
Wishlist and Compare	15	Implemented

User Profile and Account	15	Implemented
Address and Shipping	15	Implemented
Checkout and Payment	15	Implemented
Customer Support and Information	15	Implemented

3.4 Service-Wise Test Cases

Authentication Tests (15)

Test ID	Scenario	Spec File
AUTH-001	Login page opens with correct title and heading	auth-001-login-page-opens-with-correct-title-and-heading.spec.js
AUTH-002	Login page shows email and password fields	auth-002-login-page-shows-email-and-password-fields.spec.js
AUTH-003	Login page shows login button	auth-003-login-page-shows-login-button.spec.js
AUTH-004	Login page shows remember me and forgot password options	auth-004-login-page-shows-remember-me-and-forgot-password-options.spec.js
AUTH-005	Login page shows registration option for new customer	auth-005-login-page-shows-registration-option-for-new-customer.spec.js
AUTH-006	Password recovery page opens correctly	auth-006-password-recovery-page-opens-correctly.spec.js
AUTH-007	Password recovery page has email field and recover button	auth-007-password-recovery-page-has-email-field-and-recover-button.spec.js
AUTH-008	Register page opens correctly	auth-008-register-page-opens-correctly.spec.js
AUTH-009	Register page shows gender choices	auth-009-register-page-shows-gender-choices.spec.js
AUTH-010	Register page shows personal detail fields	auth-010-register-page-shows-personal-detail-fields.spec.js
AUTH-011	Register page shows password fields	auth-011-register-page-shows-password-fields.spec.js
AUTH-012	Register button is available	auth-012-register-button-is-available.spec.js
AUTH-013	Protected account page redirects guest user to login	auth-013-protected-account-page-redirects-guest-user-to-login.spec.js

AUTH-014	Protected orders page redirects guest user to login	auth-014-protected-orders-page-redirects-guest-user-to-login.spec.js
AUTH-015	Header exposes login and register links	auth-015-header-exposes-login-and-register-links.spec.js

Product and Search Tests (15)

Test ID	Scenario	Spec File
PROD-001	Home page shows welcome content and featured products	prod-001-home-page-shows-welcome-content-and-featured-products.spec.js
PROD-002	Top navigation exposes catalog categories	prod-002-top-navigation-exposes-catalog-categories.spec.js
PROD-003	Books category shows product cards	prod-003-books-category-shows-product-cards.spec.js
PROD-004	Computers category shows subcategories	prod-004-computers-category-shows-subcategories.spec.js
PROD-005	Electronics category shows subcategories	prod-005-electronics-category-shows-subcategories.spec.js
PROD-006	Apparel and shoes category shows products	prod-006-apparel-and-shoes-category-shows-products.spec.js
PROD-007	Digital downloads category opens	prod-007-digital-downloads-category-opens.spec.js
PROD-008	Jewelry category opens with product cards	prod-008-jewelry-category-opens-with-product-cards.spec.js
PROD-009	Gift cards category opens with products	prod-009-gift-cards-category-opens-with-products.spec.js
PROD-010	Build your own computer product details are visible	prod-010-build-your-own-computer-product-details-are-visible.spec.js
PROD-011	Laptop product details are visible	prod-011-laptop-product-details-are-visible.spec.js
PROD-012	Smartphone product details are visible	prod-012-smartphone-product-details-are-visible.spec.js
PROD-013	Book product page shows price and shipping text	prod-013-book-product-page-shows-price-and-shipping-text.spec.js
PROD-014	Category page has sort and page size controls	prod-014-category-page-has-sort-and-page-size-controls.spec.js
PROD-015	Search page returns computer products	prod-015-search-page-returns-computer-products.spec.js

Cart Tests (15)

Test ID	Scenario	Spec File
CART-001	Empty cart page opens	cart-001-empty-cart-page-opens.spec.js
CART-002	Empty cart message is shown	cart-002-empty-cart-message-is-shown.spec.js
CART-003	Header shopping cart link is visible	cart-003-header-shopping-cart-link-is-visible.spec.js
CART-004	Cart footer link is visible	cart-004-cart-footer-link-is-visible.spec.js
CART-005	Book product has add to cart controls	cart-005-book-product-has-add-to-cart-controls.spec.js
CART-006	Laptop product has add to cart controls	cart-006-laptop-product-has-add-to-cart-controls.spec.js
CART-007	Computer product has add to cart controls	cart-007-computer-product-has-add-to-cart-controls.spec.js
CART-008	Gift card product has add to cart controls	cart-008-gift-card-product-has-add-to-cart-controls.spec.js
CART-009	Apparel product has cart controls	cart-009-apparel-product-has-cart-controls.spec.js
CART-010	Jewelry product has cart controls	cart-010-jewelry-product-has-cart-controls.spec.js
CART-011	Cart page still shows search controls	cart-011-cart-page-still-shows-search-controls.spec.js
CART-012	Cart page shows order progress steps	cart-012-cart-page-shows-order-progress-steps.spec.js
CART-013	Cart page shows information footer links	cart-013-cart-page-shows-information-footer-links.spec.js
CART-014	Cart page keeps footer follow links available	cart-014-cart-page-keeps-footer-follow-links-available.spec.js
CART-015	Cart page has checkout journey labels	cart-015-cart-page-has-checkout-journey-labels.spec.js

Wishlist and Compare Tests (15)

Test ID	Scenario	Spec File
WISH-001	Wishlist page opens	wish-001-wishlist-page-opens.spec.js
WISH-002	Empty wishlist message is shown	wish-002-empty-wishlist-message-is-

		shown.spec.js
WISH-003	Header wishlist link is visible	wish-003-header-wishlist-link-is-visible.spec.js
WISH-004	Gift card product has wishlist button	wish-004-gift-card-product-has-wishlist-button.spec.js
WISH-005	Computer product has compare button	wish-005-computer-product-has-compare-button.spec.js
WISH-006	Laptop product has compare button	wish-006-laptop-product-has-compare-button.spec.js
WISH-007	Compare products page opens	wish-007-compare-products-page-opens.spec.js
WISH-008	Recently viewed products page opens	wish-008-recently-viewed-products-page-opens.spec.js
WISH-009	New products page opens	wish-009-new-products-page-opens.spec.js
WISH-010	Product page exposes email a friend action	wish-010-product-page-exposes-email-a-friend-action.spec.js
WISH-011	Email a friend page opens	wish-011-email-a-friend-page-opens.spec.js
WISH-012	Email a friend form has friend email field	wish-012-email-a-friend-form-has-friend-email-field.spec.js
WISH-013	Email a friend form has message and send button	wish-013-email-a-friend-form-has-message-and-send-button.spec.js
WISH-014	Compare products footer link is available	wish-014-compare-products-footer-link-is-available.spec.js
WISH-015	Wishlist footer link is available	wish-015-wishlist-footer-link-is-available.spec.js

User Profile and Account Tests (15)

Test ID	Scenario	Spec File
USER-001	My account page requires login for guest	user-001-my-account-page-requires-login-for-guest.spec.js
USER-002	Order history page requires login for guest	user-002-order-history-page-requires-login-for-guest.spec.js
USER-003	Addresses page requires login for guest	user-003-addresses-page-requires-login-for-guest.spec.js
USER-004	Login page supports returning customer section	user-004-login-page-supports-returning-customer-section.spec.js

USER-005	Footer exposes my account link	user-005-footer-exposes-my-account-link.spec.js
USER-006	Footer exposes orders link	user-006-footer-exposes-orders-link.spec.js
USER-007	Footer exposes addresses link	user-007-footer-exposes-addresses-link.spec.js
USER-008	Register page shows account creation form	user-008-register-page-shows-account-creation-form.spec.js
USER-009	Register page shows personal details fields	user-009-register-page-shows-personal-details-fields.spec.js
USER-010	Register page shows email field	user-010-register-page-shows-email-field.spec.js
USER-011	Register page shows password setup fields	user-011-register-page-shows-password-setup-fields.spec.js
USER-012	Login page provides register button for new user	user-012-login-page-provides-register-button-for-new-user.spec.js
USER-013	Login page provides forgot password link	user-013-login-page-provides-forgot-password-link.spec.js
USER-014	Password recovery page supports account recovery	user-014-password-recovery-page-supports-account-recovery.spec.js
USER-015	Sitemap includes account links	user-015-sitemap-includes-account-links.spec.js

Address and Shipping Tests (15)

Test ID	Scenario	Spec File
ADDR-001	Shipping returns page opens	addr-001-shipping-returns-page-opens.spec.js
ADDR-002	Shipping returns page has customer service context	addr-002-shipping-returns-page-has-customer-service-context.spec.js
ADDR-003	Guest addresses page redirects to login	addr-003-guest-addresses-page-redirects-to-login.spec.js
ADDR-004	Book product displays free shipping	addr-004-book-product-displays-free-shipping.spec.js
ADDR-005	Science product displays free shipping	addr-005-science-product-displays-free-shipping.spec.js
ADDR-006	Health book product displays delivery date	addr-006-health-book-product-displays-delivery-date.spec.js

ADDR-007	Laptop product shows in stock availability	addr-007-laptop-product-shows-in-stock-availability.spec.js
ADDR-008	Smartphone product shows in stock availability	addr-008-smartphone-product-shows-in-stock-availability.spec.js
ADDR-009	Blue jeans product shows in stock availability	addr-009-blue-jeans-product-shows-in-stock-availability.spec.js
ADDR-010	Shopping cart shows address step	addr-010-shopping-cart-shows-address-step.spec.js
ADDR-011	Shopping cart shows shipping step	addr-011-shopping-cart-shows-shipping-step.spec.js
ADDR-012	Footer has shipping returns link	addr-012-footer-has-shipping-returns-link.spec.js
ADDR-013	Sitemap includes shipping returns link	addr-013-sitemap-includes-shipping-returns-link.spec.js
ADDR-014	Cart page has checkout progress section	addr-014-cart-page-has-checkout-progress-section.spec.js
ADDR-015	Shipping page keeps information links available	addr-015-shipping-page-keeps-information-links-available.spec.js

Checkout and Payment Tests (15)

Test ID	Scenario	Spec File
PAY-001	Empty cart keeps user on shopping cart page	pay-001-empty-cart-keeps-user-on-shopping-cart-page.spec.js
PAY-002	Empty cart blocks checkout with empty cart message	pay-002-empty-cart-blocks-checkout-with-empty-cart-message.spec.js
PAY-003	Cart page is available before checkout	pay-003-cart-page-is-available-before-checkout.spec.js
PAY-004	Cart page shows payment step	pay-004-cart-page-shows-payment-step.spec.js
PAY-005	Cart page shows confirm step	pay-005-cart-page-shows-confirm-step.spec.js
PAY-006	Cart page shows complete step	pay-006-cart-page-shows-complete-step.spec.js
PAY-007	Gift card product has recipient name field	pay-007-gift-card-product-has-recipient-name-field.spec.js
PAY-008	Gift card product has recipient email field	pay-008-gift-card-product-has-recipient-email-field.spec.js

PAY-009	Gift card product has sender fields	pay-009-gift-card-product-has-sender-fields.spec.js
PAY-010	Gift card product has message field	pay-010-gift-card-product-has-message-field.spec.js
PAY-011	Gift card product has add to cart button	pay-011-gift-card-product-has-add-to-cart-button.spec.js
PAY-012	Gift card product shows fixed price	pay-012-gift-card-product-shows-fixed-price.spec.js
PAY-013	Book product shows price information	pay-013-book-product-shows-price-information.spec.js
PAY-014	Laptop product shows price information	pay-014-laptop-product-shows-price-information.spec.js
PAY-015	Smartphone product shows price information	pay-015-smartphone-product-shows-price-information.spec.js

Customer Support and Information Tests (15)

Test ID	Scenario	Spec File
INFO-001	Contact us page opens	info-001-contact-us-page-opens.spec.js
INFO-002	Contact us page has full name field	info-002-contact-us-page-has-full-name-field.spec.js
INFO-003	Contact us page has email field	info-003-contact-us-page-has-email-field.spec.js
INFO-004	Contact us page has enquiry textarea	info-004-contact-us-page-has-enquiry-textarea.spec.js
INFO-005	Contact us page has submit button	info-005-contact-us-page-has-submit-button.spec.js
INFO-006	Sitemap page opens	info-006-sitemap-page-opens.spec.js
INFO-007	Privacy notice page opens	info-007-privacy-notice-page-opens.spec.js
INFO-008	Conditions of use page opens	info-008-conditions-of-use-page-opens.spec.js
INFO-009	About us page opens	info-009-about-us-page-opens.spec.js
INFO-010	News page opens	info-010-news-page-opens.spec.js
INFO-011	Blog page opens	info-011-blog-page-opens.spec.js
INFO-012	Footer information links are visible	info-012-footer-information-links-are-visible.spec.js

INFO-013	Footer customer service links are visible	info-013-footer-customer-service-links-are-visible.spec.js
INFO-014	Newsletter subscription controls are visible	info-014-newsletter-subscription-controls-are-visible.spec.js
INFO-015	Footer social links are visible	info-015-footer-social-links-are-visible.spec.js

Milestone 4 - CI/CD and Allure Reporting

4.1 GitHub Actions Pipeline

The CI/CD workflow is stored in `.github/workflows/allure-report.yml`. It installs dependencies, installs Playwright browsers, runs the full suite, validates Allure result counts, generates the report, and pushes the static report to the `gh-pages` branch.

1. Download repository source for the current commit.
2. Run `npm ci` to install locked dependencies.
3. Install Playwright browsers with system dependencies.
4. Run `npx playwright test` for the complete test suite.
5. Verify that exactly 360 Allure result files are present and all are passed.
6. Generate the Allure HTML report.
7. Deploy the generated report to GitHub Pages.

4.2 Allure Report Hosting

The generated report is hosted through GitHub Pages from the `gh-pages` branch. The current deployed report link is:

https://ajaybhatnagar1712.github.io/Wipro_SDET_Playwright_Capstone/

Milestone 5 - Final Execution Result

5.1 Final Result Summary

Metric	Value
Total unique test cases	120
Browser projects	3
Total cross-browser executions	360
Passed	360

Failed	0
Broken	0
Skipped	0

5.2 Browser-Wise Result

Browser	Total Executions	Passed	Failed
Chromium	120	120	0
Firefox	120	120	0
WebKit	120	120	0

5.3 Final Quality Summary

The final execution achieved 100% pass rate for the implemented functional coverage. The suite validates the required 8 services with 15 test cases per service and produces a combined Allure report for review. The report also confirms browser coverage across Chromium, Firefox, and WebKit.

Conclusion

The capstone project successfully demonstrates service-wise Playwright automation using JavaScript, Page Object Model design, Allure reporting, and GitHub Pages hosting. The final project satisfies the minimum requirement of 120 test cases and provides a deployed evidence report with 360 passed cross-browser executions.

Appendix - How to Run Locally

```
cd "D:\WIPRO\Capstone_Project"
npm install
npx playwright test
npx allure generate allure-results --clean -o allure-report
npx allure open allure-report
```